MODUL 8

JURNAL

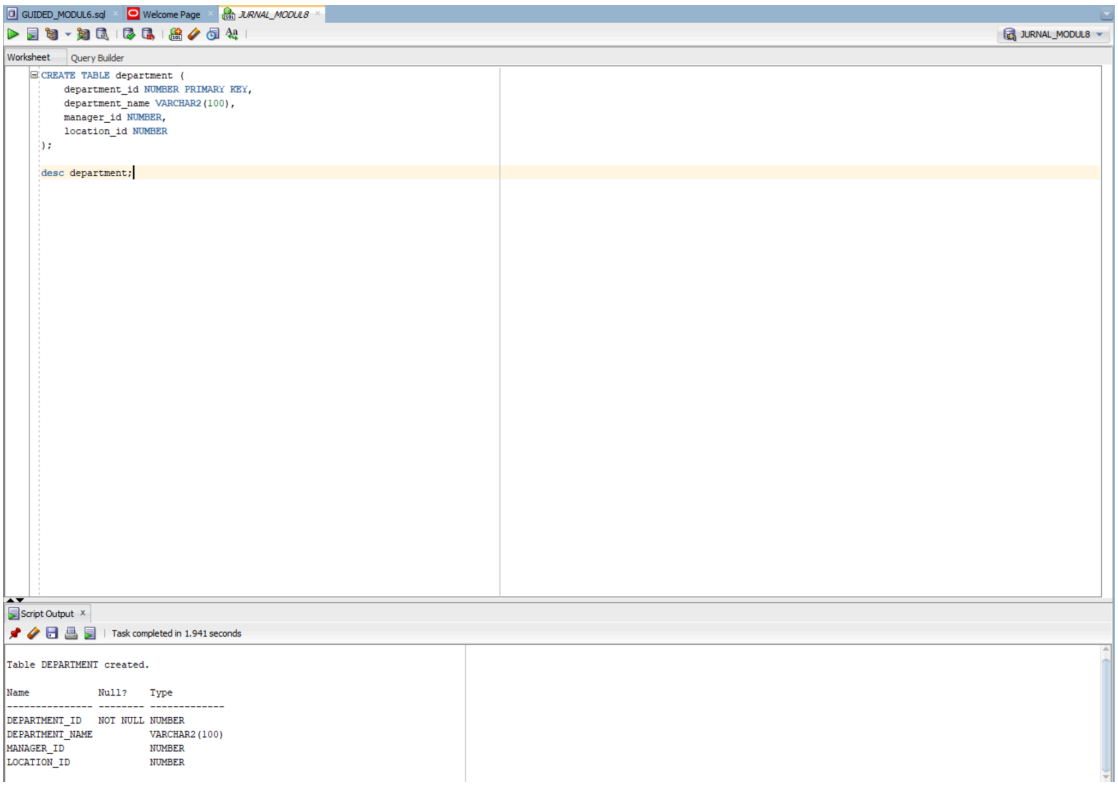
Kerjakan studi kasus pada modul 8 (halaman 87). Sertakan rekaman layar untuk setiap nomor.

8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

JAWABAN

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.



   b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL.

```
CREATE TABLE employee (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(100) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER,
    commission_pct NUMBER,
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_dept FOREIGN KEY (department_id)
    REFERENCES department(department_id)
);

desc employee;
```

```
Task completed in 0.665 seconds

Name           Null?    Type
-------------- -------- -------------
EMPLOYEE_ID    NOT NULL NUMBER
LAST_NAME      NOT NULL VARCHAR2(100)
EMAIL                   VARCHAR2(100)
SALARY                  NUMBER
COMMISSION_PCT          NUMBER
HIRE_DATE      NOT NULL DATE
DEPARTMENT_ID           NUMBER
```

c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

```
Task completed in 0.647 seconds

HIRE_DATE      NOT NULL DATE
DEPARTMENT_ID           NUMBER

Table SUPPLIER created.

Name           Null?    Type
-------------- -------- -------------
SUPPLIER_ID    NOT NULL NUMBER
SUPPLIER_NAME           VARCHAR2(100)
```

```
CREATE TABLE supplier (
    supplier_id NUMBER PRIMARY KEY,
    supplier_name VARCHAR2(100)
);

CREATE TABLE product (
    product_id NUMBER PRIMARY KEY,
    product_name VARCHAR2(100),
    supplier_id NUMBER,
    CONSTRAINT fk_supplier FOREIGN KEY (supplier_id)
    REFERENCES supplier(supplier_id)
);

desc product;
```

```
Task completed in 0.622 seconds

SUPPLIER_NAME           VARCHAR2(100)

Table PRODUCT created.

Name           Null?    Type
-------------- -------- -------------
PRODUCT_ID     NOT NULL NUMBER
PRODUCT_NAME            VARCHAR2(100)
SUPPLIER_ID             NUMBER
```

2. Analisis table-table dengan compute statistics

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE product COMPUTE STATISTICS;
```

```
Table PRODUCT created.

Name         Null?    Type
------------ -------- -------------
PRODUCT_ID   NOT NULL NUMBER
PRODUCT_NAME          VARCHAR2(100)
SUPPLIER_ID           NUMBER

Table DEPARTMENT analyzed.


Table EMPLOYEE analyzed.


Table SUPPLIER analyzed.


Table PRODUCT analyzed.
```

3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

```
INSERT INTO department VALUES (10, 'HR', 101, 1001);
INSERT INTO department VALUES (20, 'IT', 102, 1002);
INSERT INTO department VALUES (30, 'Finance', 103, 1003);
INSERT INTO department VALUES (40, 'Marketing', 104, 1004);
INSERT INTO department VALUES (50, 'Sales', 105, 1005);
INSERT INTO department VALUES (60, 'Logistic', 106, 1006);
INSERT INTO department VALUES (70, 'Admin', 107, 1007);
INSERT INTO department VALUES (80, 'Support', 108, 1008);
INSERT INTO department VALUES (90, 'RsD', 109, 1009);
INSERT INTO department VALUES (100, 'Legal', 110, 1010);
```

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

```
INSERT INTO employee VALUES (1, 'Andi', 'andi@mail.com', 5000, NULL, SYSDATE, 80);
INSERT INTO employee VALUES (2, 'Budi', 'budi@mail.com', 6000, NULL, SYSDATE, 80);
INSERT INTO employee VALUES (3, 'Citra', 'citra@mail.com', 5500, NULL, SYSDATE, 20);
INSERT INTO employee VALUES (4, 'Dedi', 'dedi@mail.com', 7000, NULL, SYSDATE, 30);
INSERT INTO employee VALUES (5, 'Eka', 'eka@mail.com', 6500, NULL, SYSDATE, 40);
INSERT INTO employee VALUES (6, 'Fajar', 'fajar@mail.com', 4800, NULL, SYSDATE, 50);
INSERT INTO employee VALUES (7, 'Gina', 'gina@mail.com', 7200, NULL, SYSDATE, 60);
INSERT INTO employee VALUES (8, 'Hadi', 'hadi@mail.com', 5300, NULL, SYSDATE, 70);
INSERT INTO employee VALUES (9, 'Intan', 'intan@mail.com', 5100, NULL, SYSDATE, 80);
INSERT INTO employee VALUES (10, 'Joko', 'joko@mail.com', 5800, NULL, SYSDATE, 90);
```

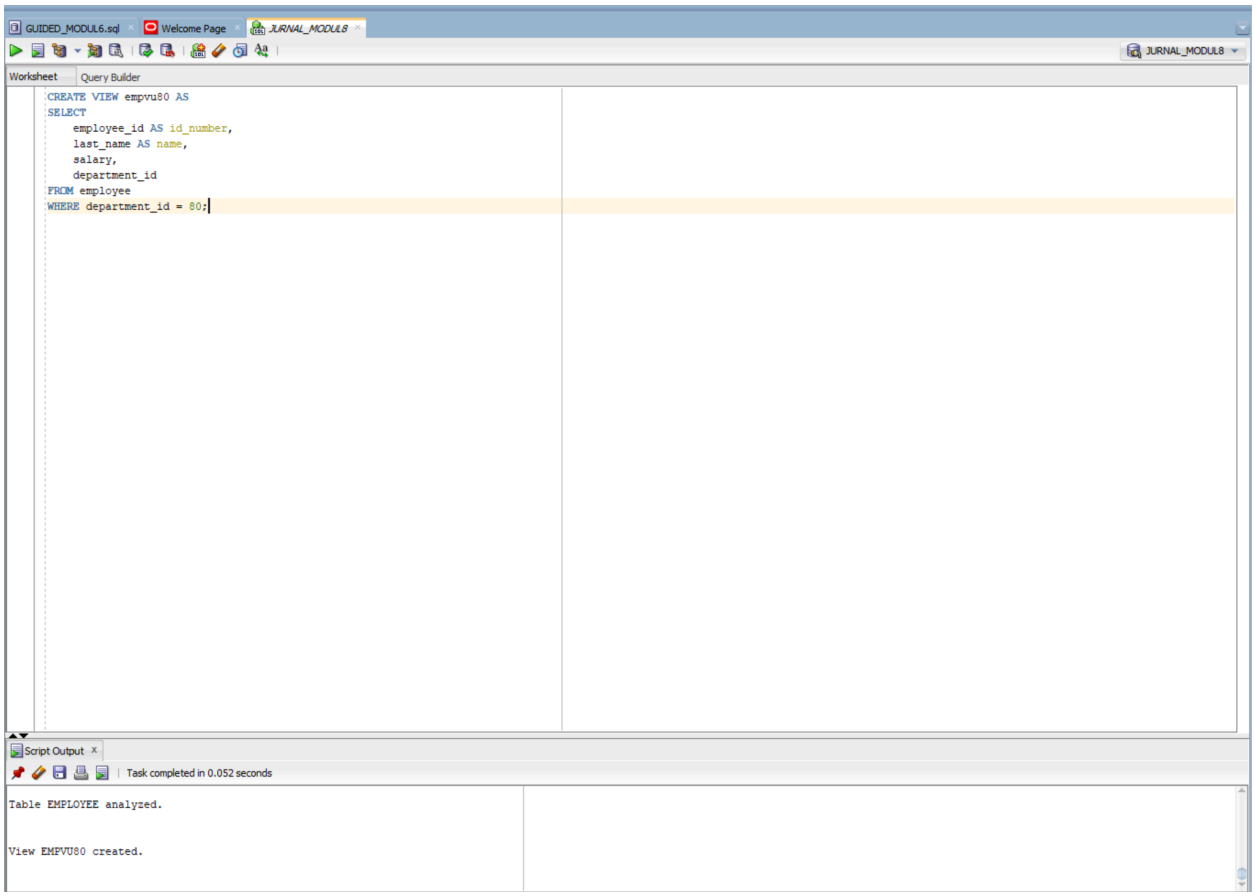
```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

4. Analisis table-table dengan compute statistics

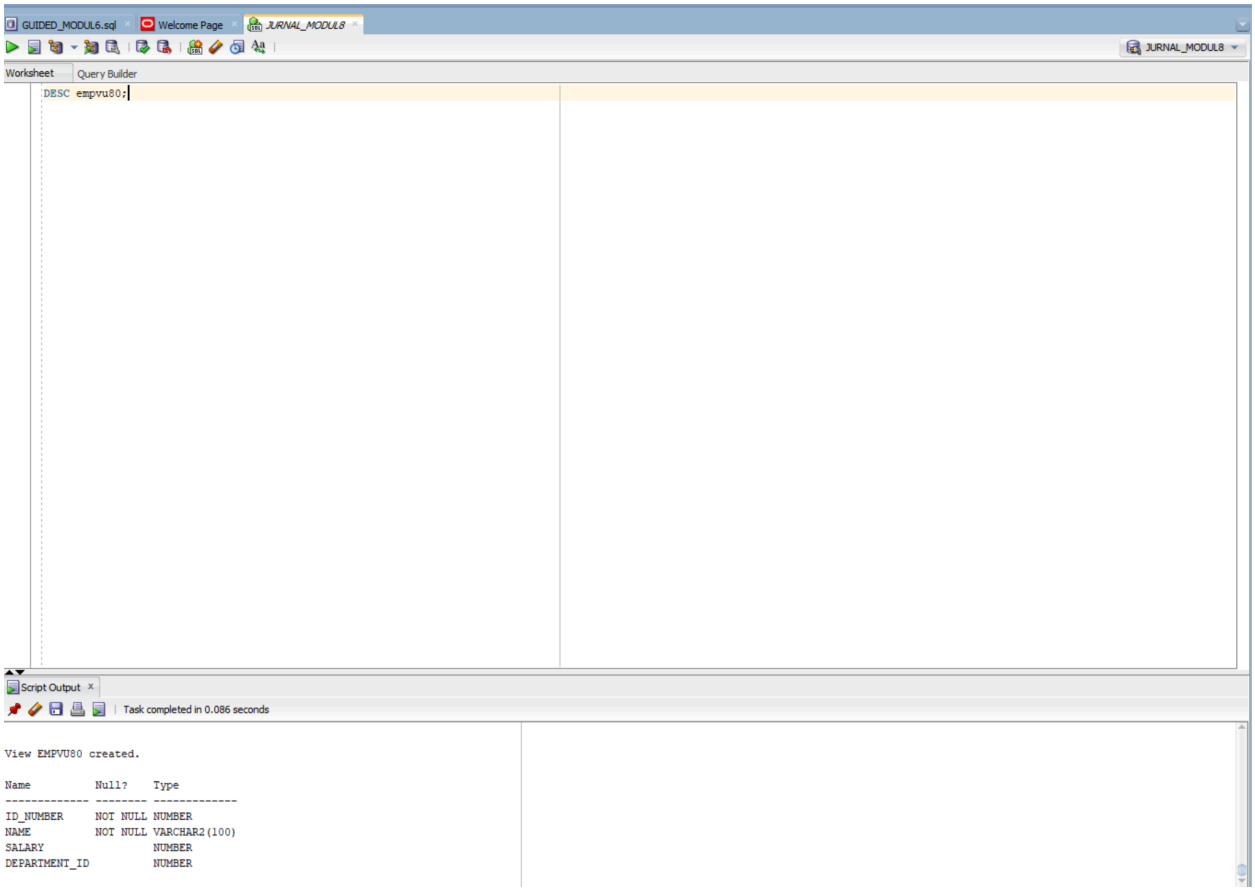
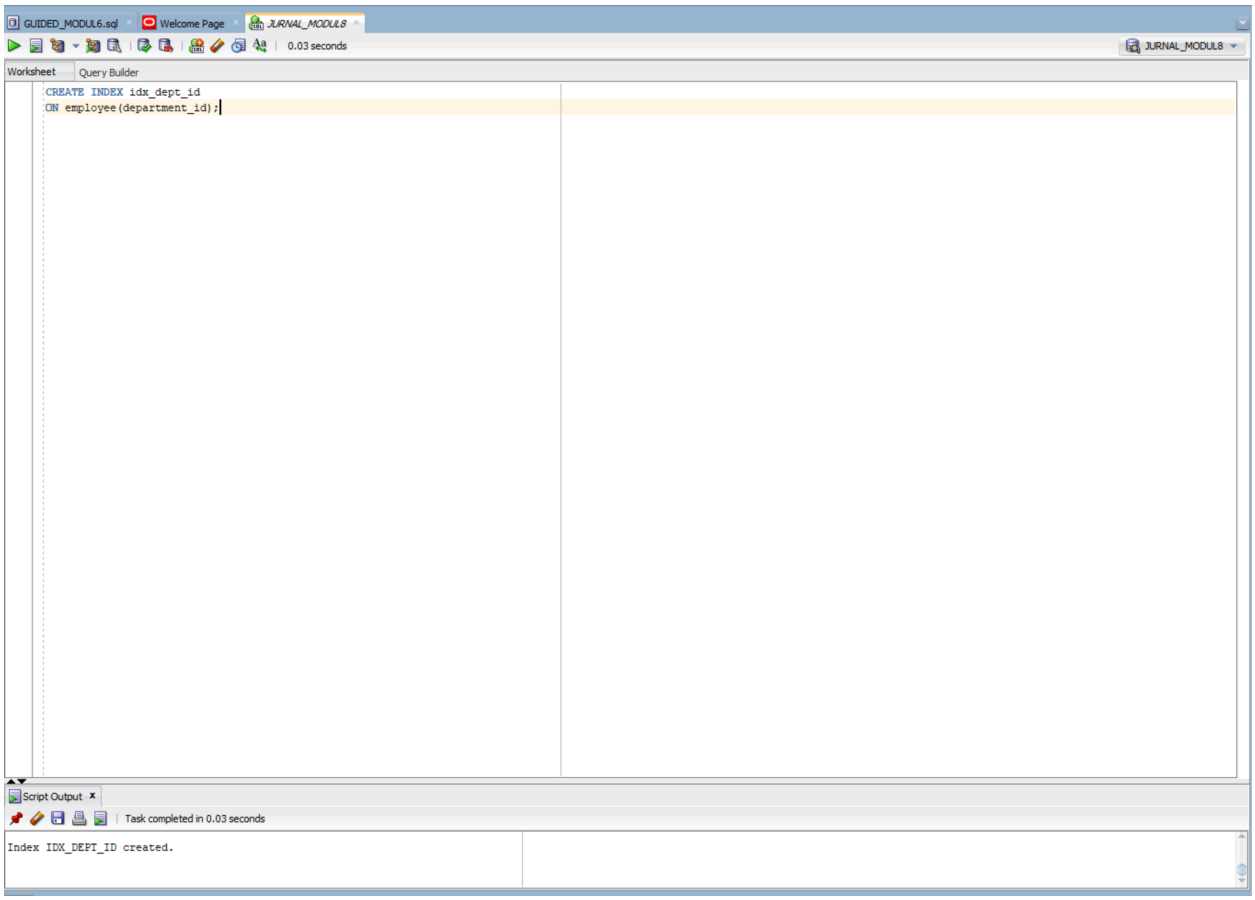
```
ANALYZE TABLE employee COMPUTE STATISTICS;
```

```
Table EMPLOYEE analyzed.
```

5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.



6. Kemudian tampilkan struktur dari view empvu80.



7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.